INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 9

1. Experiment Title:

Aliases and CASE

2. Aim:

Rename result columns, classify students as Pass or Fail using CASE, and calculate marks with bonus marks.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

An alias gives a temporary name to a column or expression in query output. CASE evaluates conditions and returns a value for the first matching condition. Arithmetic expressions can be selected as calculated columns.

5. Algorithm / Procedure:

1. Create exam_result and insert the four names and marks.
2. Use AS to rename name and marks in the output.
3. Use CASE to show Pass when marks are at least 40, otherwise Fail.
4. Add 5 to marks and name the calculated column with_bonus.

6. Program / Queries / Commands:

```
CREATE TABLE exam_result (
    name VARCHAR(20),
    marks INT
);

INSERT INTO exam_result (name, marks)
VALUES
    ('Asha', 92),
    ('Rohan', 47),
    ('Neha', 68),
    ('Meera', 33);

-- Task 1
SELECT name AS student_name, marks AS score
FROM exam_result;

-- Task 2
SELECT name,
       CASE
           WHEN marks >= 40 THEN 'Pass'
           ELSE 'Fail'
       END AS result
FROM exam_result;

-- Task 3
SELECT name, marks + 5 AS with_bonus
FROM exam_result;
```

7. Input:

Asha—92, Rohan—47, Neha—68, Meera—33.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: student_name	score
Asha	92
Rohan	47
Neha	68
Meera	33
Task 2: name	result
Asha	Pass
Rohan	Pass
Neha	Pass
Meera	Fail
Task 3: name	with_bonus
Asha	97
Rohan	52
Neha	73
Meera	38

9. Result:

The SQL statements for Experiment 9 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is a column alias?
2. What is the purpose of AS?
3. What does CASE do?
4. What happens in the ELSE branch?